

A Project Report
On
Atmospheric Water Generator

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Birla Institute of Technology and Science-Pilani,
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Certificate

This is to certify that the project report entitled “**Atmospheric Water Generator**” submitted by Mr. Aditya Varma (ID No. 2022A4PS1491H) in partial fulfillment of the requirements of the course BIO F266, Study Project Course, embodies the work done by him under my supervision and guidance.

Date:

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ABSTRACT

This MATLAB programme has been developed in the framework of a design project for an "Atmospheric Water Generator" (AWG) device, which extracts water from humid ambient air by cooling it below its dew point. In order to optimize the performance of the AWG, the precise calculation and analysis of psychrometric properties are essential. Such a program will enable one to calculate a complete set of psychrometric properties conditioned by three fundamental inputs: height above sea level, in order to make possible the definition of total atmospheric pressure, and two others like temperature, humidity ratio, or relative humidity.

In addition to its ability to return numerical results, the program offers powerful graphical capabilities. It can plot constant property curves on a psychrometric chart that will be used to understand how the various air parameters interact at different conditions. The program also allows plotting of 3D curves based on one psychrometric property to intuitively recognize any trends or correlations. Features such as these are most beneficial in emulating varied environmental conditions to optimize the design and operational efficiency of the AWG.

The system can also be used as a versatile input configuration tool: flexible and easy to use and therefore very useful for researchers and engineers in the field of air conditioning, dehumidification, and water extraction technology. Through computing and visualizing accurately the properties of atmospheric air, this MATLAB tool is part of the development and fine-tuning process of the AWG device.

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Chapter-1

Psychrometric Properties

Psychrometric properties (or properties of moist air) are used to define the properties of air which is a mixture of dry air and water vapour. These properties are very helpful in designing air-conditioning and refrigeration systems.

Here are a few psychrometric properties-

1. Dry Bulb Temperature(Degree Celsius)
2. Specific Humidity(kg/kg)
3. Relative Humidity(%)
4. Enthalpy(kj/kg)
5. Wet Bulb Temperature(Degree Celsius)
6. Specific Volume(m³/kg)
7. Vapour Pressure(kpa)
8. Dew Point Temperature(Degree Celsius)
9. Thermodynamic Wet Bulb Temperature(Degree Celsius)
10. Total Pressure(kpa)

Dry Bulb temperature

The temperature of moist-air mixture measured by a perfectly dry sensor, such as a thermocouple or thermometer.

It is denoted by T_{DB} .

Specific Humidity

Specific Humidity or Humidity Ratio is the ratio of the mass of water vapour to the mass of dry air in the moist-air mixture.

It is denoted by W .

Relative Humidity

Relative Humidity is the ratio of the partial pressure of water vapour to the saturation pressure of water vapour at the existing Dry Bulb Temperature.

It is denoted by ϕ .

Enthalpy

Specific Enthalpy of moist air is equal to the sum of dry air enthalpy and that of water vapour.

It is denoted by h .

Wet Bulb Temperature

The temperature of a thermometer with a wetted wick over which air flows at a specific velocity.

It is denoted by T_{WB} .

Specific Volume

Specific Volume is the volume of the mixture per unit mass of dry air.

It is denoted by v .

Vapour Pressure

The partial pressure of water vapour in a homogeneous air-water vapour mixture is the pressure exerted by water vapour alone.

Saturation Pressure at Dew Point Temperature.

It is denoted by p_{vap} .

Dew Point Temperature

The temperature at which a moist-air mixture at a given humidity ratio becomes saturated.

It is denoted by T_{DP} .

Thermodynamic Wet Bulb Temperature

Almost same as Wet Bulb Temperature but this time the temperature drop of wetted wick uses Adiabatic process.

It is denoted by T_{WBT} .

Total Pressure

The pressure of Atmospheric Air at an altitude from sea level.

It is denoted by p_{tot} .

Chapter-2

Relations between psychrometric properties

Symbols used:

1. Dry Bulb Temperature-DB
2. Specific Humidity-SH
3. Relative Humidity-RH
4. Enthalpy-ET
5. Wet Bulb Temperature-WB
6. Specific Volume-SV
7. Vapour Pressure-PV
8. Dew Point Temperature-DP
9. Total Pressure-PT
10. Height above sea level-H

Side Quantities:

1. Saturation pressure at DB-PS
2. Saturation pressure at WB-PSWET

Calculating PT when H is given

$$PT = 101.325 * ((1 - (2.25577 * (10^{(-5)}) * H))^{5.2559}) \text{ (upto 11000m)}$$

Where H is in metres

Saturation Pressure(P_{sat}) when Temperature(T) is given

$$P_C=22100$$

$$T_C=647.30$$

$$K=4.39553-6.2442*((T+273.15)/1000)+9.953*(((T+273.15)/1000)^2)-5.151*(((T+273.15)/1000)^3)$$

$$P_{sat}=P_C*(10^{(K*(1-(T_C/(T+273.15))))})$$

Saturation Temperature(T_{sat}) when pressure(P) is given

$$ALPHA=\log(P)$$

$$T_{sat}=6.54+14.526*ALPHA+0.7389*(ALPHA^2)+0.0949*(ALPHA^3)+0.4569*(P^{0.1984}) \text{ (T is between 0 Degree Celsius and 93 Degree Celsius)}$$

Enthalpy when Dry Bulb Temperature and Specific Humidity are given

$$ET=(1.005*DB)+SH*(2500+(1.88*DB))$$

Specific Humidity when Vapour Pressure is given

$$SH=0.622*(P_V/(P_T-P_V))$$

Relative Humidity when Dry Bulb Temperature and Vapour are given

$$RH=(P_V/P_S)*100$$

Specific Volume when Dry Bulb Temperature and Vapour Pressure are given

$$SV=0.2871*(DB+273.15)/(P_T-P_V)$$

Wet Bulb Temperature when Dry Bulb Temperature and Relative Humidity are given

WB=-

$$4.391976+(0.0198197*RH)+(0.526359*DB)+(0.00730271*RH*DB)+(2.4315*(10^{(-4)})*(RH^{(2)}))-(-2.58101*(10^{(-5)})*DB*(RH^{(2)}))$$

Relative Humidity when Dry Bulb Temperature and Wet Bulb Temperature are given

$$P=PT*((DB-WB)/1514)*(1+(WB/873))$$

$$RH=((PSWET-P)/PS)*100$$

Chapter-3

Psychrometric Calculator

The approach I have used for the Psychrometric calculator program is when three Psychrometric properties are given, it calculates all other Psychrometric properties. The given properties include H(which calculates PT) and two other properties which are not mutually dependent.

Calculation of some properties require Iterative analysis.

Given below are the calculations that require Iterative Analysis:

1. DB when SH/PV/DP and WB are given

- $PC=22100$
- $TC=647.30$
- $K=4.39553-$
 $6.2442*((DB+273.15)/1000)+9.953*(((DB+273.15)/1000)^2)-$
 $5.151*(((DB+273.15)/1000)^3)$
- $PS=PC*(10^{(K*(1-(TC/(DB+273.15))))})$
- $KWET=4.39553-$
 $6.2442*((WB+273.15)/1000)+9.953*(((WB+273.15)/1000)^2)-$
 $5.151*(((WB+273.15)/1000)^3)$
- $PSWET=PC*(10^{(KWET*(1-(TC/(WB+273.15))))})$
- $P=PT*((DB-WB)/1514)*(1+(WB/873))$

- $RH = ((PSWET - P) / PS) * 100$
2. DB when RH and ET are given
- $RH = (PV / PS) * 100$
 - $ET = (1.005 * DB) + SH * (2500 + (1.88 * DB))$
 - $SH = 0.622 * (PV / (PT - PV))$
3. DB when RH and SV are given
- $RH = (PV / PS) * 100$
 - $SV = 0.2871 * (DB + 273.15) / (PT - PV)$
 - $PC = 22100$
 - $TC = 647.30$
 - $K = 4.39553 - 6.2442 * ((DB + 273.15) / 1000) + 9.953 * (((DB + 273.15) / 1000)^2) - 5.151 * (((DB + 273.15) / 1000)^3)$
 - $PS = PC * (10^{(K * (1 - (TC / (DB + 273.15))))})$
4. DB when ET and SV are given
- $ET = (1.005 * DB) + SH * (2500 + (1.88 * DB))$
 - $SV = 0.2871 * (DB + 273.15) / (PT - PV)$
 - $SH = 0.622 * (PV / (PT - PV))$
5. DB when WB and SV are given
- $SV = 0.2871 * (DB + 273.15) / (PT - PV)$
 - $RH = (PV / PS) * 100$
 - $PC = 22100$
 - $TC = 647.30$
 - $K = 4.39553 - 6.2442 * ((DB + 273.15) / 1000) + 9.953 * (((DB + 273.15) / 1000)^2) - 5.151 * (((DB + 273.15) / 1000)^3)$
 - $PS = PC * (10^{(K * (1 - (TC / (DB + 273.15))))})$

Everything works fine.

The only drawbacks of the program are when it calculates the values of psychrometric properties using iterative methods and the value of properties converges to some other values.

The methods I have used for Iterative Analysis are:

1. Bisection Method
2. Newton-Raphson Method

Assumptions:

1. Calculation of RH when DB and WB are given also requires Iterative analysis but since we are considering Daikin Psychrometrics Program as a standard, we are trying to get our values as close to it as possible and there is a deviation in the values provided by Iterative analysis and Software. So, we are using an Empirical relation from the internet.
Link- <https://www.mdpi.com/2073-4433/13/11/1765>
2. SH, PV and DP are mutually dependent so, it is not possible to find other psychrometric properties when any two of these three are given.
3. It is not possible to find other psychrometric properties when both ET and WB are given because they are incompatible to define a psychrometric point.

Chapter-4

Curve Plotting

2D graph

A 2D graph of constant Psychrometric properties is plotted when a Psychrometric calculation is done. All the final psychrometric properties are considered as constants and a curve is plotted between SH and DB.

DB is taken as a vector and an equation between SH and DB is used to plot the curve.

3D Surface

A 3D surface is plotted between SH, DB and PT where DB and PT are vectors. It is used when a psychrometric property is given and we want to find variations in SH over a range of values of DB and PT.

Full Graph

A full Psychrometric chart is plotted taking DB as vector.

Chapter-5

How the program works

The program consists of many functions that does their required tasks.

1. `psychrometry()`: This is the main function of the program and provides an interactive environment for the user and asks them what they want to do. It runs in an infinite loop until it receives 'n' as input to choice for if the user wants to perform another calculation. It calls other functions for their operations.
2. `psychrometric_calculator()`: This function is called by `psychrometry()` function and it calls other functions as per the choice of the user.
3. `PP1_PP2()`: There are 28 such functions. These functions are called by `psychrometric_calculator()` function and they find all the other psychrometric properties. They call `graph2D()` function and gives all the Psychrometric properties as arguments.
4. `graph2D()`: This function takes all the Psychrometric properties as arguments and plots a 2D graph between SH and DB taking all the Psychrometric properties as constants.
5. `graph3D()`: This function plots a 3D surface and is called by `psychrometry()` function. It takes a Psychrometric property as input and plots a constant property surface between SH, DB and PT.
6. `graph_full()`: This function is called by `psychrometry()` function and plots a full Psychrometric Chart.

Chapter-6

Future plan

1. Making the program robust and eliminating all the problems occurred using Iterative analysis.
2. When finding Saturation temperature when Pressure is given, the range of Saturation temperature is between 0 Degree Celsius and 93 Degree Celsius. To make the program work for a wide range to temperatures and pressures, we have to implement ASHRAE steam tables onto the program.
3. Make the program as a GUI.

Conclusion

This development project was aimed at developing a MATLAB program to aid in optimizing and analyzing an Atmospheric Water Generator (AWG) system. The output of this program is the psychrometric properties with three inputs: usually height from sea level-total pressure but now also two other key properties such as either temperature or humidity ratio. Psychrometric property calculations are necessary in understanding and improving the performance of an AWG, especially under changing environmental conditions.

Also, the possibility of constant property curves, 3D plots, and psychrometric charts provide yet another interesting visualization tool for researchers and engineers. The software analyzes the relationship between the properties of air, thereby affording an opportunity to refine and enhance design and operational parameters of the AWG. Therefore, the MATLAB tool is highly powerful but not unnecessarily complex, thus making provision for simulation and analysis of the AWG systems and consequently leading to better decision-making and efficiency. The real-time data integration and additional features further enhance applicability towards both research and practical implementation in the near future.

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